From Intuition to Rigour: A Proposition of Quantum Temporal Discontinuity under Critical Examination

De l'Intuition à la Rigueur : Une Proposition de Discontinuité Temporelle Quantique sous Examen Critique

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Abstract

We present a three-part study of a speculative hypothesis concerning the temporal structure of quantum matter. **First**, we state the original proposition: a formal analogy between the spatially discontinuous localization of the atom (nucleus concentrated within a void) and a possible temporal discontinuity of particle existence. This intuition, conceived in the 1990s and formalized via AI assistance, postulates that particles would "flicker" in time with a characteristic scale $\tau_0 \sim 10^{-24}$ s. **Second**, we subject this proposition to systematic critical analysis, identifying four major conceptual flaws: the ontological incoherence of "temporal non-existence," the arbitrary choice of temporal scale, the apparent violation of quantum unitarity, and the confusion between mathematical discretization and physical discontinuity. **Third**, adopting a *scientific counterfactual* approach, we derive the theoretical and experimental consequences this hypothesis would generate if it were valid despite its flaws: effective violation of Lorentz symmetry, stroboscopic resonances in field theory, intrinsic decoherence mechanisms, and five alternative experimental protocols (precision spectroscopy of forbidden transitions, relativistic desynchronization of entangled states, collider resonance archaeology, limits on macroscopic superpositions, and Hawking radiation anomalies). Our work illustrates how a raw intuition, even partially defective, can generate a scientifically productive research program when subjected to AI-assisted formal rigour.

Keywords: discrete time, quantum foundations, temporal granularity, decoherence, experimental metaphysics, AI-assisted formalization, conceptual critique

Mots-clés : temps discret, fondements de la mécanique quantique, granularité temporelle, décohérence, métaphysique expérimentale, formalisation assistée par IA, critique conceptuelle

Introduction: A Three-Movement Approach

This article adopts an unusual but methodologically transparent tripartite structure. We start from a simple observation: contemporary theoretical physics produces fundamental hypotheses either by rigorous deduction from established principles or by empirical induction from experimental data. A third path, too often obscured, is that of **analogical intuition**—the recognition of structural patterns between distinct domains—followed by **critical examination** and **counterfactual exploration** of its implications.

*Independent researcher. Original intuition conceived circa 1990; formalization and critical analysis 2024 with AI assistance.

The proposition examined here illustrates this trajectory. Originally conceived in the 1990s as a raw intuition (termed by its author "quantiquo-épilepsie" or "quantum epilepsy"), it was preserved in informal notes for three decades for lack of technical apparatus for its scientific articulation. Its transformation into a formal research object was carried out in 2024 through an iterative dialogue with large language model systems, where artificial intelligence functioned as *formal translator*—converting metaphorical associations into mathematical structures, identifying implicit presuppositions, and imposing disciplinary constraints (causality, unitarity, falsifiability) that raw intuition typically ignores.

Our three-part approach reflects this epistemic progression:

I. The Statement: Faithful presentation of the original proposition, in its formalized form but prior to any critique.

II. The Examination: Systematic analysis of conceptual, methodological, and formal flaws.

III. The Exploration: Rigorous derivation of consequences, adopting the hypothesis as a *scientific counterfactual*—a classical method consisting in exploring "what would happen if" a defective proposition were nonetheless valid.

This structure aims to answer a transparency requirement: the reader must be able to evaluate the original proposition on its own terms before learning of its flaws, then understand how these flaws do not exhaust the scientific productivity of the underlying intuition.

I Statement of the Original Proposition

1 Context and Fundamental Intuition

1.1 The space-time asymmetry in atomic physics

Atomic physics establishes that baryonic matter exhibits a fundamentally discontinuous spatial structure. The standard atomic model describes a nucleus of characteristic dimensions $R_{\text{nucleus}} \sim 10^{-15}$ m, concentrating more than 99.9% of mass, surrounded by an electronic cloud extending to $R_{\text{atom}} \sim 10^{-10}$ m. This defines a spatial presence density:

$$\eta_{\text{space}} = \frac{V_{\text{nucleus}}}{V_{\text{atom}}} \sim 10^{-15} \tag{1}$$

established experimentally by Rutherford scattering (1911) and confirmed by modern spectroscopy.

Conversely, quantum mechanics postulates continuous temporal existence of quantum states. The time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \tag{2}$$

presupposes differentiable evolution with uniform action of the evolution operator $\hat{U}(t, t_0) = \exp\left[-i\hat{H}(t - t_0)/\hbar\right]$ over all temporal intervals. No fundamental temporal granularity is introduced in standard formalism.

1.2 The central analogical intuition

The original proposition rests on a simple, almost naive question: *if matter is discontinuous in space—concentrated in a tiny nucleus within an atomic void—why would it be continuous in time?*

This intuition postulates a **formal symmetry** between spatial dimensions and the temporal dimension in the localization of matter. If atomic space is "granular" (concentrated presence/diffuse absence), time could be so as well (intermittent existence/non-existence).

1.3 The temporal existence factor

To quantify this intuition, the proposition introduces a **temporal existence factor** $\mathcal{E}$, by direct analogy with the spatial density η_{space}:

Definition 1 (Temporal existence factor)**.** *Let $\tau_{existence}$ be the cumulative duration of effective particle existence over a reference period τ_{cycle}. The temporal existence factor is defined by:*

$$\mathcal{E} = \frac{\tau_{existence}}{\tau_{cycle}} \tag{3}$$

In this vision, a particle would "exist" during discrete intervals $\tau_{\text{existence}}$, separated by "gaps" of non-existence $\tau_{\text{non-existence}} = \tau_{\text{cycle}} - \tau_{\text{existence}}$, such that $\mathcal{E} < 1$.

2 Characteristic Scales and Formalization

2.1 The hydrogen atom scale

For the atomic electron, the characteristic orbital period in the hydrogen ground state is:

$$\tau_{\text{orbit}} = \frac{2\pi a_0}{v_{\text{Bohr}}} = \frac{4\pi\hbar^3}{m_e e^4} \approx 1.52 \times 10^{-16} \text{ s} \tag{4}$$

where a_0 is the Bohr radius and $v_{\text{Bohr}} = \alpha c$ the electron velocity in the Bohr model ($\alpha \approx 1/137$ the fine-structure constant).

If one postulates a formal symmetry with atomic spatial density (Eq. 1), one would obtain a scale of "non-existence":

$$\tau_{\text{non-existence}} \sim \tau_{\text{orbit}} \cdot (1 - \eta_{\text{space}}) \approx \tau_{\text{orbit}} \tag{5}$$

However, this direct analogy is physically problematic (as we shall see in Part II) because it violates causality constraints.

2.2 Proposed scale: the electroweak domain

The alternative proposition suggests an intermediate scale associated with electroweak interactions:

$$\tau_{\text{proposed}} \sim 10^{-24} \text{ s} \tag{6}$$

corresponding to the characteristic timescale of weak interactions ($\tau_W \sim \hbar/(m_W c^2) \approx 4 \times 10^{-27}$ s) or hadronic resonances.

2.3 Discrete-time Hamiltonian

The mathematical formalization proposes to modify the Schrödinger equation by treating time as a discrete variable $t_n = n \cdot \tau_0$, with τ_0 the temporal granularity scale. Evolution becomes:

$$|\psi(t_{n+1})\rangle = \hat{U}(\tau_0)\,|\psi(t_n)\rangle \tag{7}$$

with $\hat{U}(\tau_0) = e^{-i\hat{H}\tau_0/\hbar}$. In the limit $\tau_0 \to 0$, one recovers continuous Schrödinger evolution.

A leading-order expansion gives:

$$\frac{|\psi(t_{n+1})\rangle - |\psi(t_n)\rangle}{\tau_0} = -\frac{i}{\hbar}\hat{H}\,|\psi(t_n)\rangle - \frac{\tau_0}{2\hbar^2}\hat{H}^2\,|\psi(t_n)\rangle + \mathcal{O}(\tau_0^2) \tag{8}$$

The corrective term $\propto \tau_0$ would represent a deviation from standard continuous evolution, detectable if τ_0 is finite.

2.4 Modification of the time-energy uncertainty relation

The proposition suggests that the Mandelstam-Tamm uncertainty relation:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \tag{9}$$

could be modified by fundamental temporal granularity. If Δt is quantized in units of τ_0, the lower bound on energy uncertainty would become:

$$\Delta E \geq \frac{\hbar}{2\tau_0} \cdot \frac{1}{n} \quad (n \in \mathbb{N}^*) \tag{10}$$

implying a "quantization" of minimal energy uncertainty.

II Critical Analysis: Conceptual and Methodological Flaws

3 Flaw I: The Ontological Category Error

3.1 Two incommensurable types of discontinuity

The central analogy of the proposition conflates **two ontologically distinct discontinuities**:

Spatial discontinuity (established)	**Temporal discontinuity (proposed)**
Matter *localized* in reduced volume	Matter *existing intermittently*
Inter-atomic void is absence of matter	"Temporal void" is ontologically undefined
Electronic cloud occupies intermediate space	No equivalent "temporal cloud" proposed
Particle exists continuously in time	Particle ceases to exist during gaps

Table 1: Ontological asymmetry in the space-time analogy

3.2 The aporia of temporal non-existence

The critical lacuna: the proposition omits to define **what constitutes the "temporal non-existence" of a spatially existing particle**. The following questions remain unanswered:

- Does the particle cease interacting with fields during $\tau_{\text{non-existence}}$?
- Are quantum numbers (charge, spin, mass) conserved or suspended?
- Do entanglement correlations persist across "temporal holes"?
- Is the wave function defined during non-existence?

Without answers to these questions, the proposition remains more metaphor than operational physics. Spatial non-existence is trivial (absence of detection); temporal non-existence of an existing object is conceptually incoherent without additional metaphysical machinery.

4 Flaw II: The Arbitrary Choice of Scale

4.1 Absence of derivation mechanism

The proposed scale $\tau_0 \sim 10^{-24}$ s is justified by numerical proximity to weak interaction timescales ($\tau_W \sim 4 \times 10^{-27}$ s). However:

Remark 1 (Arbitrary scale coupling)**.** *No physical mechanism links atomic electron dynamics to weak interaction scales. Electrons are first-generation leptons without colour; they do not participate significantly in weak interaction (coupling is suppressed by the ratio $m_e/m_W \sim 10^{-5}$). The selection of 10^{-24} s represents a numerical coincidence without causal foundation.*

A rigorous treatment requires either:

(a) Derivation from first principles (e.g., quantum gravity)

(b) Phenomenological coupling to known physics (e.g., Planck scale effects $t_P = \sqrt{\hbar G/c^5} \approx 5.39 \times 10^{-44}$ s)

(c) Experimental constraint from existing data

None of these options is satisfied in the original proposition.

5 Flaw III: Tension with Quantum Unitarity

5.1 Discretization vs. discontinuity

The discrete evolution equation (Eq. 7):

$$|\psi(t_{n+1})\rangle = \hat{U}(\tau_0)\,|\psi(t_n)\rangle \tag{11}$$

describes **discretized continuous evolution**, not **interrupted existence**. If the particle "does not exist" between t_n and t_{n+1}, the evolution operator should be:

- Undefined (no evolution possible for a non-existent entity)
- Or projected onto identity (formal conservation without dynamics)

Yet the operator $\hat{U}(\tau_0) = \exp\left(-i\hat{H}\tau_0/\hbar\right)$ presupposes **continuous unitary propagation** over interval τ_0. The formalism conflates *mathematical discretization* with *physical discontinuity*—a distinction with distinct measurable consequences.

5.2 The conservation problem

If existence is truly intermittent, what becomes of probability conservation? Unitarity requires $\langle\psi(t)\rangle\,\psi(t) = 1$ for all t. During "non-existence" periods, should the norm drop to zero? If so, particle reappearance would violate norm conservation. If not, "non-existence" is merely a semantic redefinition of continuous evolution.

6 Flaw IV: Misapplication of Time-Energy Uncertainty

6.1 Confusion between granularity and lifetime

The argument suggesting that quantizing Δt in units of τ_0 modifies the uncertainty relation (Eq. 10):

$$\Delta E \geq \frac{\hbar}{2\tau_0} \cdot \frac{1}{n} \tag{12}$$

is formally incorrect. The Mandelstam-Tamm relation (Eq. 9) concerns the **lifetime** of quantum states—the characteristic time of significant state change—not fundamental temporal granularity.

Quantizing Δt would imply that excited state lifetimes are strict integer multiples of τ_0. This is a testable prediction (absent from observed spectroscopic linewidths, which show continuous distribution), distinct from the claimed "quantization of energy uncertainty."

6.2 Incoherence with standard quantum mechanics

Moreover, the relation $\Delta E \cdot \Delta t \geq \hbar/2$ is not an uncertainty relation in the Robertson sense (like $\Delta x \cdot \Delta p \geq \hbar/2$), as time is not a Hermitian operator in standard quantum mechanics. The proposition confuses uncertainty in measurement time with an intrinsic property of time itself.

7 Assessment of Critical Analysis

The original proposition, though intuitively appealing, suffers from grave conceptual flaws:

1. **Ontological incoherence**: "Temporal non-existence" is not operationally defined.
2. **Scale arbitrariness**: The choice of $\tau_0 \sim 10^{-24}$ s lacks physical foundation.
3. **Unitarity violation**: The formalism presupposes disguised continuous evolution.
4. **Formal error**: Misinterpretation of the time-energy uncertainty relation.

These flaws do not mean the intuition is devoid of value. They indicate that the proposition, as it stands, does not constitute a coherent physical theory. However, adopting a *scientific counterfactual* approach, we can explore the consequences this hypothesis would generate if it were valid despite its defects—this is the object of Part III.

III Theoretical and Experimental Consequences (Counterfactual Exploration)

8 Working Hypothesis and Methodology

Let us admit, for the developments that follow, that the temporal discontinuity hypothesis is valid despite the flaws identified in Part II. We adopt the following specifications:

- A granularity scale $\tau_0 \sim 10^{-24}$ s exists fundamentally.
- Particles exist by "pulses" of duration $\tau_{\text{existence}}$, separated by $\tau_{\text{non-existence}}$ gaps.
- Evolution between pulses is unitary; "non-existence" is treated as a projection phase (non-dynamical).

This counterfactual exploration aims to derive empirical predictions which, if observed, could signal real temporal granularity, independently of the defective original formulation.

9 Theoretical Consequences

9.1 Modification of quantum field propagators

The Feynman propagator $D_F(x-y)$ integrates over all continuous temporal paths. With fundamental granularity τ_0, the discrete sum gives:

$$D_F^{\text{disc}}(x-y) = \sum_{n=0}^{\infty} \int \frac{d^3p}{(2\pi)^3} \frac{i}{2E_{\mathbf{p}}} \left(e^{-iE_{\mathbf{p}}(t_x - t_y - n\tau_0)} \theta(t_x - t_y - n\tau_0) + \text{h.c.} \right) e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \tag{13}$$

Proposition 1 (Stroboscopic resonances)**.** *For scattering processes with energy transfer ΔE satisfying $\Delta E \cdot \tau_0 = 2\pi k$ ($k \in \mathbb{Z}$), constructive interferences between temporal paths produce* ***resonance peaks*** *in cross-sections at energies:*

$$E_k = \frac{2\pi k\hbar}{\tau_0} \approx k \times 4.1 \ GeV \quad (for\ \tau_0 = 10^{-24}\ s) \tag{14}$$

These resonances would be narrow ($\Gamma \sim \hbar/\tau_0 \sim 0.4$ eV) and devoid of decay channels (as they are not real particles but temporal granularity artifacts).

9.2 Effective violation of Lorentz symmetry

A privileged temporal scale τ_0 breaks Lorentz invariance (as does any privileged frame or scale). The dispersion relation modifies to:

$$E^2 = p^2c^2 + m^2c^4 + \delta E(\tau_0, p), \quad \delta E \sim \frac{\hbar}{\tau_0} \sin^2\left(\frac{E\tau_0}{2\hbar}\right) \tag{15}$$

implying:

- Energy-dependent group velocities $v_g = dE/dp$ deviating from c for ultra-relativistic particles.
- Modified pair production thresholds.
- Vacuum birefringence effects for high-energy photons.

9.3 Intrinsic decoherence mechanism

For a macroscopic superposition of N particles, desynchronization of individual "temporal phases" $\phi_i(t) = 2\pi t/\tau_0^{(i)}$ (where $\tau_0^{(i)}$ may vary due to local gravitational potential or velocity) yields a decoherence rate:

$$\Gamma_{\text{gran}} \sim N \cdot \frac{(\delta\tau_0)^2}{\tau_0^3} \tag{16}$$

This predicts **fundamental decoherence** independent of environmental coupling, increasing with system size.

Conjecture 1 (Synchronization requirement)**.** *Observation of macroscopic superpositions (SQUIDs, molecular interferometry) implies either (a) $\tau_0 \ll 10^{-24}$ s, or (b) a* ***synchronization mechanism*** *enforcing $\tau_0^{(i)} = \tau_0^{(j)}$ for bound systems. The latter suggests gravity-mediated temporal coordination, falsifiable via superposition experiments in variable gravitational potentials.*

10 Experimental Protocols

Direct temporal resolution at $\tau_0 \sim 10^{-24}$ s exceeds current attosecond technology ($\sim 10^{-18}$ s) by six orders of magnitude. We propose five alternative protocols:

10.1 Protocol I: Precision spectroscopy of forbidden transitions

Mechanism: If existence is intermittent, "forbidden" transitions (violating selection rules) become partially allowed by "temporal tunneling"—the particle bypassing continuous evolution constraints during non-existence intervals.

Implementation:

- Target: Forbidden transitions (e.g., $\Delta l = 0$ for electric dipole transition) in trapped ions ($^{40}Ca^+$, $^{171}Yb^+$)
- Observable: Residual linewidth $\Gamma_{\text{forbidden}}$ vs. high-precision QED predictions
- Sensitivity: Atomic fountain spectroscopy reaches $\Delta\nu/\nu \sim 10^{-18}$; predicted effect at $\tau_0 \cdot \nu_{\text{transition}} \sim 10^{-9}$ for optical transitions ($\nu \sim 10^{15}$ Hz)

Signature: Systematic excess linewidth increasing with transition multipolarity, unexplainable by standard QED effects.

10.2 Protocol II: Relativistic desynchronization in entanglement

Mechanism: Entangled particles in relative motion undergo differential time dilation of their "granularity frequencies" $\nu = 1/\tau_0$:

$$\Delta\nu = \nu_0 \left(\sqrt{1 - \frac{v_1^2}{c^2}} - \sqrt{1 - \frac{v_2^2}{c^2}} \right) \approx \frac{\nu_0}{2c^2}(v_2^2 - v_1^2) \tag{17}$$

For atomic velocities $v \sim 10^3$ m/s, $\Delta\nu/\nu_0 \sim 10^{-12}$.

Implementation:

- Create EPR pairs (photons or atoms)
- Separate into trajectories with controlled velocity difference $\Delta v \sim 10^2$–10^3 m/s
- Measure Bell inequality violation visibility vs. Δv

Signature: Quadratic visibility degradation with Δv, absent in standard quantum mechanics.

10.3 Protocol III: Collider resonance archaeology

Mechanism: Stroboscopic resonances (Proposition 1) appear as periodic peaks at $E_n = n \cdot 2\pi\hbar/\tau_0$.

Implementation:

- Analyze existing LHC, Belle II, BaBar data for narrow peaks at $E_n = n \times 4.1$ GeV ($n \in \mathbb{Z}^+$)
- Criteria: width $\Gamma \sim 0.4$ eV, absence of decay channels, non-assignability to Standard Model

Signature: Periodic structure in effective mass spectrum, incompatible with known hadrons but consistent with $\tau_0 = 10^{-24}$ s.

10.4 Protocol IV: Limits on macroscopic superpositions

Mechanism: Desynchronization decoherence (Eq. 16) imposes a fundamental limit on the size of quantum superpositions.

Implementation:

- Extend molecular interferometry (current record: 10^4 amu) to masses 10^6–10^9 amu (clusters, nanoparticles)
- Rigorously control environmental decoherence (cryogenic temperature, ultra-high vacuum)
- Monitor coherence time τ_{coh} vs. number of constituents N

Signature: $\tau_{\text{coh}} \propto N^{-2}$ (granularity prediction) vs. N^{-1} (environmental decoherence) vs. constant (standard QM). An abrupt coherence drop beyond a threshold N_{crit} would be revealing.

10.5 Protocol V: Hawking radiation anomalies

Mechanism: The Hawking radiation characteristic time $\tau_H = 8\pi GM/(c^3)$ corresponds to temperature $T_H = \hbar c^3/(8\pi GMk_B)$. If $\tau_0 > \tau_H$, temporal granularity suppresses radiation.

Implementation:

- Search for primordial black holes of mass $M \sim c^3\tau_0/(8\pi G) \sim 10^8$ kg (Schwarzschild radius $\sim 10^{-19}$ m)
- Detection by gravitational microlensing (OGLE, Gaia, Fermi)
- Verify absence of characteristic Hawking gamma-ray spectrum

Signature: Compact objects of 10^8 kg, detectable by lensing but electromagnetically silent, distinguishable from radiating primordial black holes of different mass.

11 Discussion: From Flaw to Scientific Productivity

11.1 The epistemology of constructive failure

The four flaws identified in Part II do not condemn the original intuition to oblivion. They transform it into a **research program** in the methodology of Lakatos: the "hard core" (space-time symmetry of localization) is protected by a "protective belt" of modifiable auxiliary hypotheses.

For example:

- The ontological flaw can be resolved by reformulation in terms of **temporal degrees of freedom** rather than binary existence.
- Scale arbitrariness can be constrained by **experimental limits** (Protocols I–V).
- Tension with unitarity can be lifted by a **projection theory** during gaps.

11.2 The role of AI in constructive criticism

The present analysis illustrates an advanced use of artificial intelligence in theoretical physics: not as hypothesis generator (the "hard core" remains human), but as **systematic formal critic**. AI enabled:

1. Identification of type inadequacies (discretization vs. discontinuity)
2. Formalization of implicit contradictions (unitarity vs. non-existence)
3. Generation of testable predictions from a defective proposition

This "critical logical compiler" function represents a novel AI modality in science, complementary to computational and synthetic roles.

Conclusion

We have conducted a three-movement study of a three-decade-old physical intuition. **First**, we stated the temporal discontinuity proposition in its formalized form, highlighting its central intuition: the symmetry between spatial and temporal localization of matter. **Second**, we subjected this proposition to systematic critique, revealing four major conceptual flaws preventing it from constituting a coherent physical theory as it stands. **Third**, adopting a counterfactual posture, we derived the theoretical and experimental consequences this hypothesis would generate if valid, culminating in five empirical test protocols.

This approach illustrates a possible path for contemporary fundamental physics: the use of artificial intelligence as mediator between human conceptual imagination and formal rigour, enabling even defective intuitions to generate productive research programs. Whether nature respects this particular intuition—whether time is granular like space—remains an open question. But the capacity to transform a "quantiquo-épilepsie" into five verifiable experimental protocols demonstrates the power of critical formalization, human and artificial combined.

Acknowledgments

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